

Discounted Economic Profit Equals Discounted Free Cash Flow

This appendix demonstrates algebraically the equivalence between discounted cash flow and discounted economic profit. In the first section, we convert the key value driver formula presented in Chapter 3 into a value driver formula based on economic profit. This formula is used in Chapter 10 to estimate continuing value in the economic-profit valuation. The second section of this appendix generalizes the proof to any set of cash flows.

PROOF USING PERPETUITIES

To convert the key value driver formula into an economic-profit-based formula, start with the growing cash flow perpetuity:

$$V = \frac{FCF_{t=1}}{WACC - g}$$

where

V = value of operations

$FCF_{t=1}$ = free cash flow in year 1

WACC = weighted average cost of capital

g = growth in NOPAT

In Chapter 3, we convert the growing perpetuity into the key value driver formula:

$$V = \frac{NOPAT_{t=1} \left(1 - \frac{g}{RONIC} \right)}{WACC - g}$$

where $\text{NOPAT}_{t=1}$ = net operating profit after taxes

RONIC = return on new invested capital

The key value driver formula can be rearranged further into a formula based on economic profit. We do this to demonstrate that discounted cash flow is equivalent to the book value of invested capital plus the present value of future economic profit.

To begin, start with the key value driver formula, and replace NOPAT with invested capital times return on invested capital (ROIC):

$$V = \frac{\text{Invested Capital}_0 \times \text{ROIC} \times \left(1 - \frac{g}{\text{RONIC}}\right)}{\text{WACC} - g}$$

If we assume that the return on new invested capital (RONIC) equals the return on existing invested capital (ROIC), it is possible to simplify the preceding equation by distributing ROIC in the numerator:¹

$$V = \text{Invested Capital}_0 \left(\frac{\text{ROIC} - g}{\text{WACC} - g} \right)$$

To complete the transformation to economic profit, add and subtract WACC in the numerator:

$$V = \text{Invested Capital}_0 \left(\frac{\text{ROIC} - \text{WACC} + \text{WACC} - g}{\text{WACC} - g} \right)$$

Separate the fraction into two components, and then simplify:

$$\begin{aligned} V &= \text{Invested Capital}_0 \left(\frac{\text{ROIC} - \text{WACC}}{\text{WACC} - g} \right) + \text{Invested Capital}_0 \left(\frac{\text{WACC} - g}{\text{WACC} - g} \right) \\ &= \text{Invested Capital}_0 + \text{Invested Capital}_0 \left(\frac{\text{ROIC} - \text{WACC}}{\text{WACC} - g} \right) \end{aligned}$$

¹ This equation highlights two requirements for using the key value driver formula: both WACC and ROIC must be greater than the rate of growth in cash flows. If WACC is less than the cash flow growth rate, cash flows grow faster than they can be discounted, and value approaches infinity. (Perpetuity-based formulas should never be used to value cash flows whose growth rates exceed WACC.) If ROIC is lower than the growth rate, cash flows are negative, producing a negative value. In actuality, this situation is unlikely; investors would not finance a company that is never expected to generate or enable positive cash flow.

Economic profit is defined as invested capital times the difference of ROIC minus WACC. Substituting this definition into the previous equation leads to our final equation:

$$V = \text{Invested Capital}_0 + \frac{\text{Economic Profit}_1}{\text{WACC} - g}$$

According to this formula, a company's operating value equals the book value of its invested capital plus the present value of all future economic profits. (The final term is a growing perpetuity of economic profits.) If future economic profits are expected to be zero, the intrinsic value of a company equals its book value. In addition, if future economic profits are expected to be less than zero, then enterprise value should trade at less than the book value of invested capital—an occurrence observed in practice.

GENERALIZED PROOF

The previous section limited our proof to a set of cash flows growing at a constant rate. This section generalizes the proof to any set of cash flows. To demonstrate equivalence, start by computing the present value of a periodic stream of cash flows:

$$V = \sum_{t=1}^{\infty} \frac{\text{FCF}_t}{(1 + \text{WACC})^t}$$

where V = value of operations

FCF_t = free cash flow in year t

WACC = weighted average cost of capital

To this value, add and subtract the cumulative sum of all current and future amounts of invested capital (IC):

$$V = \sum_{t=0}^{\infty} \frac{\text{IC}_t}{(1 + \text{WACC})^t} - \sum_{t=0}^{\infty} \frac{\text{IC}_t}{(1 + \text{WACC})^t} + \sum_{t=1}^{\infty} \frac{\text{FCF}_t}{(1 + \text{WACC})^t}$$

where IC_t = invested capital for year t .

Next, adjust the preceding equation slightly to restate the same value using terms that can be canceled later. First, strip invested capital at time zero from the first cumulative sum. Then modify the second cumulative sum to $t = 1$ to infinity, by changing each t inside the second cumulative sum to $t - 1$. This

new representation is identical to the original representation but will allow us to cancel terms later. The new representation is as follows:

$$V = IC_0 + \sum_{t=1}^{\infty} \frac{IC_t}{(1+WACC)^t} - \sum_{t=1}^{\infty} \frac{IC_{t-1}}{(1+WACC)^{t-1}} + \sum_{t=1}^{\infty} \frac{FCF_t}{(1+WACC)^t}$$

Multiply the second cumulative sum by $(1+WACC)/(1+WACC)$. This action converts the exponent $t-1$ in the denominator of the cumulative sum to t . Also substitute for free cash flow in the third cumulative sum, using its definition, NOPAT less the increase in invested capital:

$$V = IC_0 + \sum_{t=1}^{\infty} \frac{IC_t}{(1+WACC)^t} - \sum_{t=1}^{\infty} \frac{(1+WACC)IC_{t-1}}{(1+WACC)^t} + \sum_{t=1}^{\infty} \frac{NOPAT_t - (IC_t - IC_{t-1})}{(1+WACC)^t}$$

Because there is now a consistent denominator across all three cumulative sums, combine them into a single cumulative sum:

$$V = IC_0 + \sum_{t=1}^{\infty} \frac{IC_t - (1+WACC)IC_{t-1} + NOPAT_t - IC_t + IC_{t-1}}{(1+WACC)^t}$$

In the second term of the numerator, distribute $(1+WACC)IC_{t-1}$ into its two components, IC_{t-1} and $WACC(IC_{t-1})$:

$$V = IC_0 + \sum_{t=1}^{\infty} \frac{IC_t - IC_{t-1} - WACC(IC_{t-1}) + NOPAT_t - IC_t + IC_{t-1}}{(1+WACC)^t}$$

Simplify by collecting terms:

$$V = IC_0 + \sum_{t=1}^{\infty} \frac{NOPAT_t - WACC(IC_{t-1})}{(1+WACC)^t}$$

The numerator is the definition of economic profit, so the result is a valuation based on economic profit:

$$V = IC_0 + \sum_{t=1}^{\infty} \frac{\text{Economic Profit}_t}{(1+WACC)^t}$$

The enterprise value of a company equals the book value of its invested capital plus the present value of all future economic profits. To calculate the value correctly, you must calculate economic profit using last year's (i.e., beginning-of-year) invested capital—a subtle but important distinction.

The interdependence of invested capital, economic profit, and free cash flow is not surprising. Think of discounted cash flow this way: a portion of future cash flows is required to cover the required return for the investor's capital. The remaining cash flow is either used to grow invested capital (to generate additional future cash flows) or returned to investors as an extra bonus. This bonus is valuable, so investors are willing to pay a premium for cash flows above the amount required. Subsequently, companies with positive economic profits will trade at a premium to the book value of invested capital.

